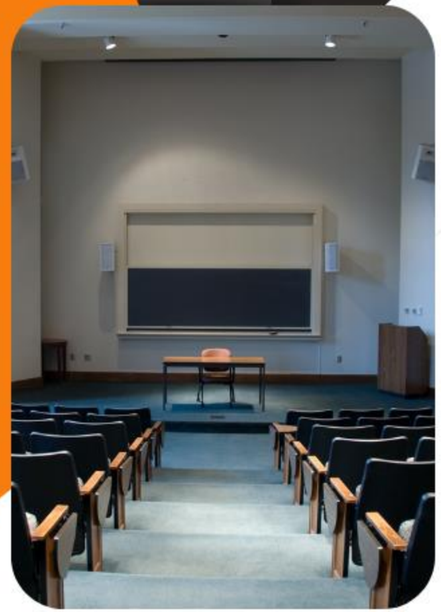
Future Skills
Business Decision Making with Interpreting Data - MCBI61

2025

Aoraki University

Student Services



Presented by **Emily Cardeno**

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Aoraki University

Student Services Dept

- Aoraki University, established in 1998, is a mid-sized university located on New Zealand's South Island.
- Aoraki offers a wide range of undergraduate and postgraduate programmes across areas such as business, science, arts, and technology.

- To better support its students, by offering academic support, career advice, mental health resources, and wellness programmes.

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Business Scenario

And Mission

- Low engagement rates in programs
- Did not find the services relevant to their real-life challenges
- Need for a data-driven understanding of student lifestyles

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Business Questions

General Objective

Increase overall student engagement by offering services that better match students' current needs, habits, and challenges. To understand student behaviour to tailor offerings.

Questions

- Which life factors affect academic performance and mental health?
- Which demographic segments would benefit from academic or wellness products?
- Are students with part-time jobs or lower attendance more likely to need tutoring?

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Project Process Overview

Data Acquisition and Preparation

- Data collection
- Data cleaning and transformation
- Data wrangling
- Data validation

Exploratory Data Analysis

- Identify patterns and behaviour found in the data that contributes to the main objective

Diagnostic Analysis

- Carry out diagnostic analysis and interpret the results
- Interpret the results

Recommendation

- Provide recommendations
- Product proposal

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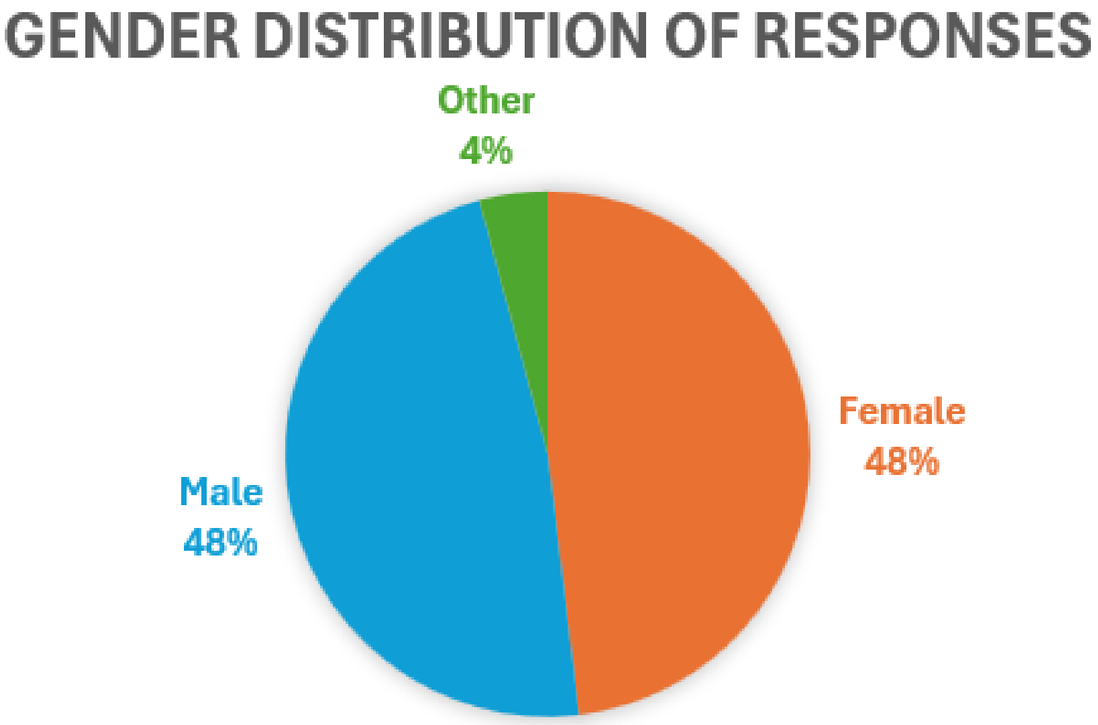
Findings

Exploratory Data Analysis



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Findings

Exploratory Data Analysis

Lifestyle Trends by Gender

- Hours of sleep
- Study hours
- Screen time
- Diet quality
- Mental health self check-in
- Exercise frequency
- Students with part time jobs

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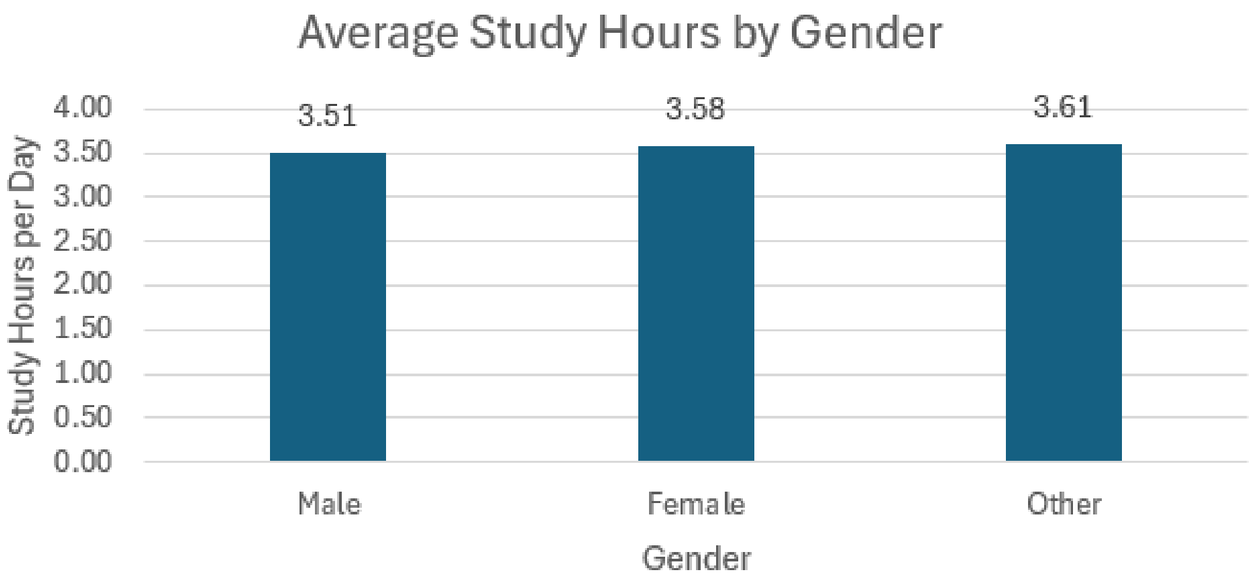
Findings

Exploratory Data Analysis

Lifestyle Trends by Gender



The chart shows that female students spend slightly more time studying per day (3.58 hrs) than male students (3.51 hrs), with students identifying as 'Other' averaging 3.61 hrs. This suggests possible gender-related differences in academic habits, which student services could explore when designing support programmes.

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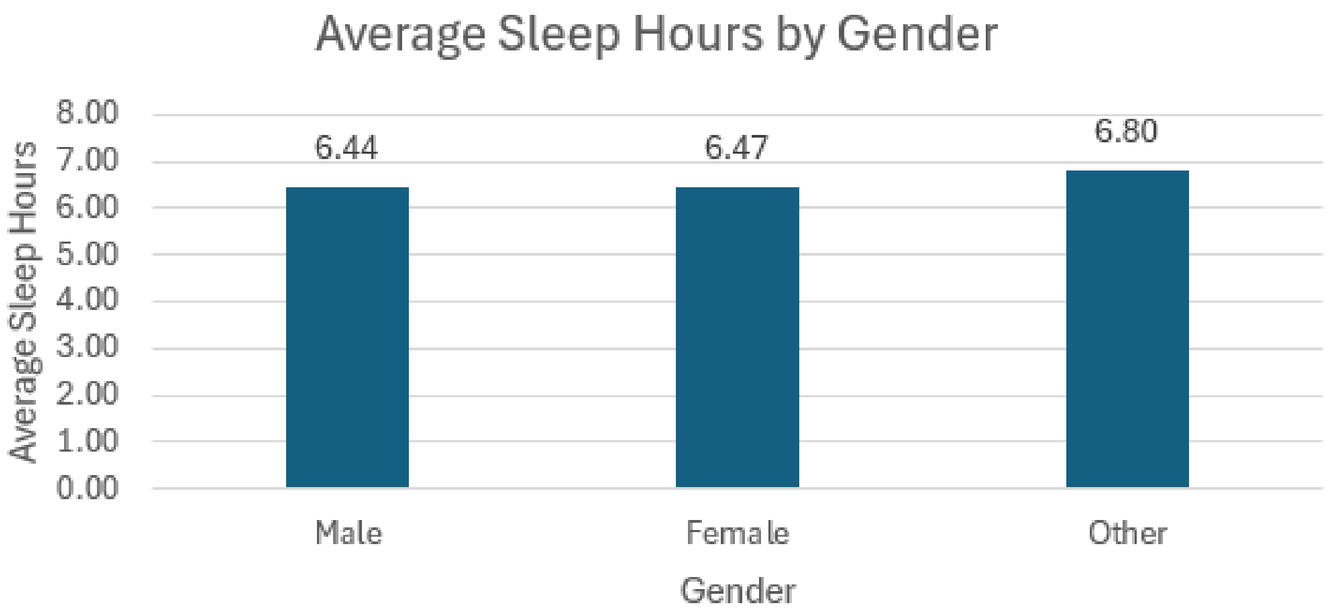
Findings

Exploratory Data Analysis

Lifestyle Trends by Gender



The chart shows there is minimal difference in the sleep habits between male and female. The students identifying as “Other” have a slightly higher average. However, the average for all genders falls below the recommended number of sleep hours (8 hours).

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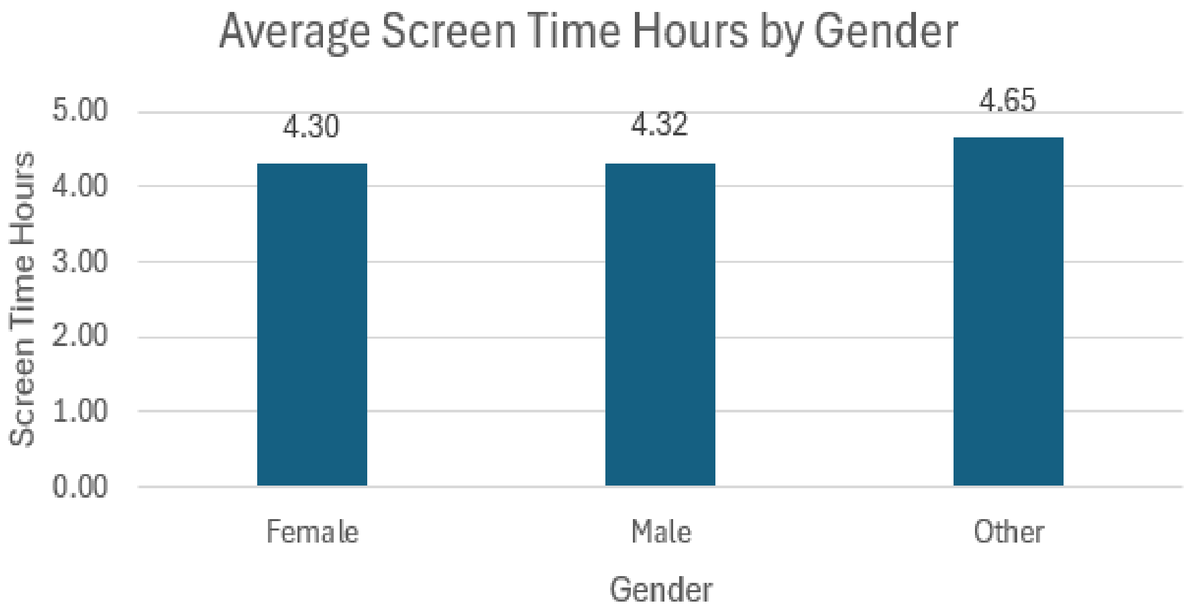
Findings

Exploratory Data Analysis

Lifestyle Trends by Gender

Netflix + Social Media = Total Screen time



The chart shows that the average screen time for all genders is more than 4 hours per day. Females slightly consumes less than males, while those identifying as “Other” have a higher average of screen time.

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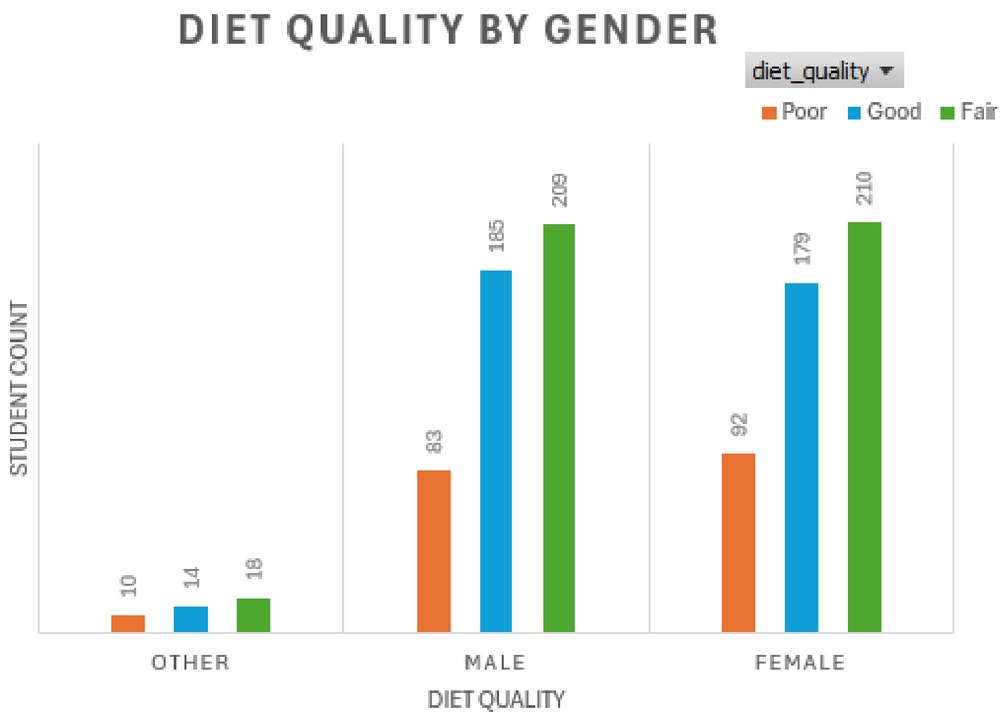
Findings

Exploratory Data Analysis

Lifestyle Trends by Gender



The graph shows the how many males, females and other genders fall under the Poor, Fair and Good Diet Categories. For all genders the highest count is under the Fair Diet Category.

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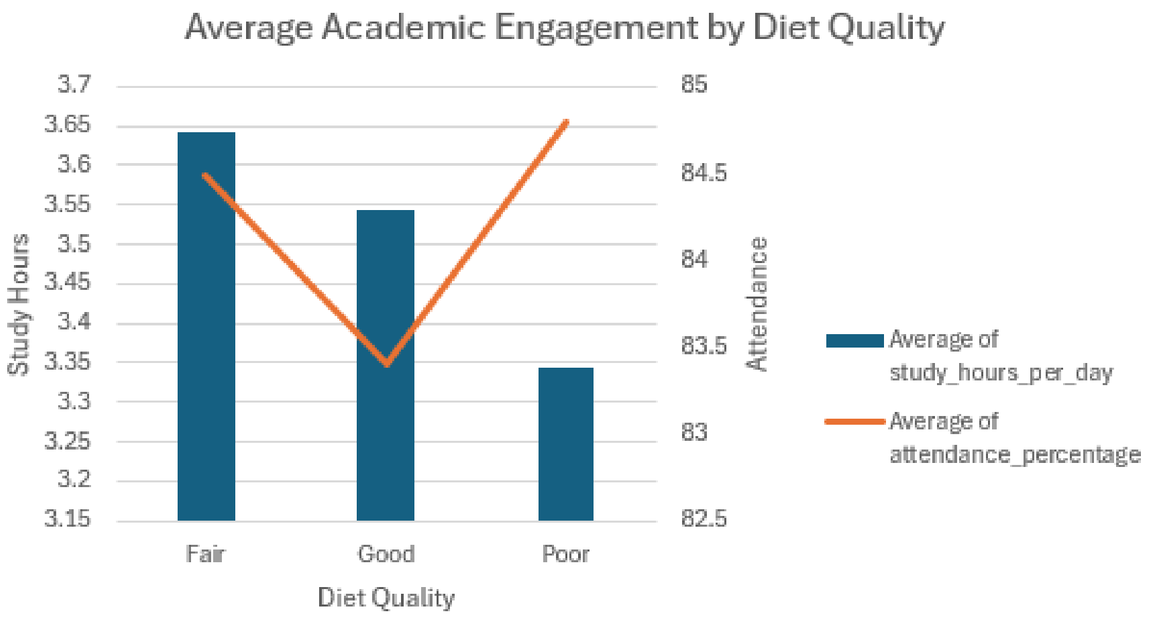
Exploratory Data Analysis

Lifestyle Trends by Gender

The graph shows that Fair diet group studies the most (about 3.65 hours/day). This is slightly counterintuitive to what we might expect regarding better diets will mean higher study hours, but here "Fair" leads, not "Good".

Poor diet group has the highest attendance (84.8%), while Fair diet group is slightly lower (84.2%). Again, this challenges assumptions — better diet does not correspond with higher attendance.



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Findings

Diagnostic Analysis

Part-time Job Impact on Study Hours

P-value = 0.3574 is **greater than 0.05.** Therefore, we **fail to reject** the null hypothesis.

There is **no statistically significant difference** in average daily study hours between students with and without part-time jobs in this sample.

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Study Hours-No	785.0000	2804.4000	3.5725	2.0826
Study Hours-Yes	215.0000	745.7000	3.4684	2.4340

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	1.8294	1.0000	1.8294	0.8477	0.3574	3.8508
Within Groups	2153.6506	998.0000	2.1580			
Total	2155.4800	999.0000				

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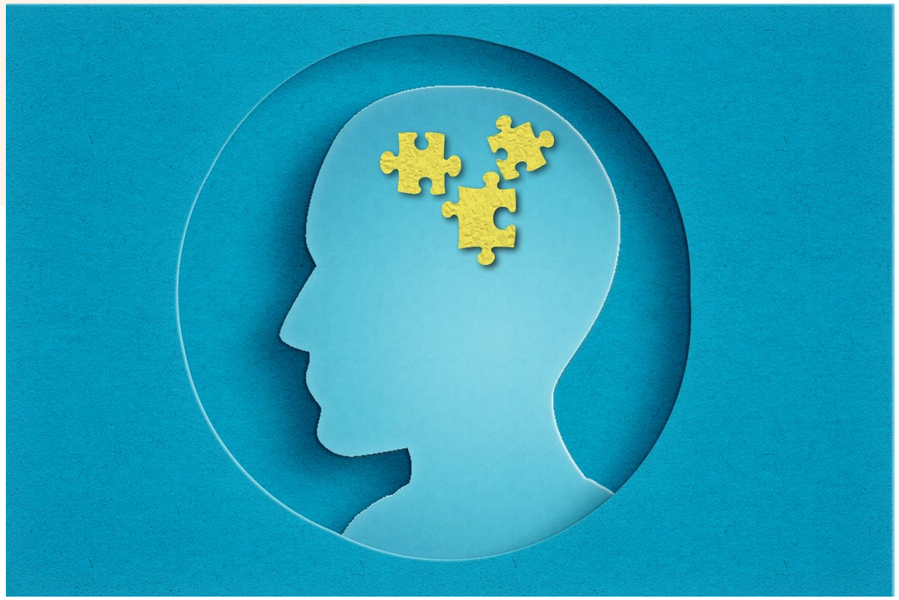
Presentation of
Insights



All genders reported over 4 hours of daily screen time

Lifestyle factors such as diet may not predict academic behaviour

Sleeping trend among students may indicate a wellbeing concern

Employment alone does not directly reduce academic effort

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Conclusion

Gender Lifestyle Trends

Wellness and academic products or services should be targeted not just by demographic group, but by behaviour-based segmentation. For instance, students with poor sleep-to-study balance could benefit from time management coaching, focus strategies, or wellness workshops.

Working Students

While it is commonly assumed that part-time work limits academic performance, the analysis found no significant difference in study time between students with and without jobs. Low attendance was more consistently linked with lower study hours and lifestyle imbalances

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Student Wellness & Engagement Tracker

85 % Attendance

5.41 Mental Health

6.43 Hours Sleep

3.01 Exercise/Wk

239 Extracurricular Participation

School Year

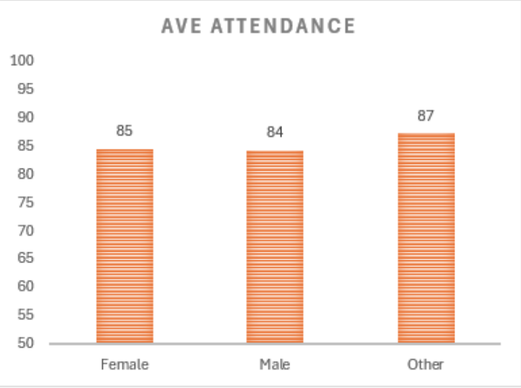
2022 | 2021 | 2020

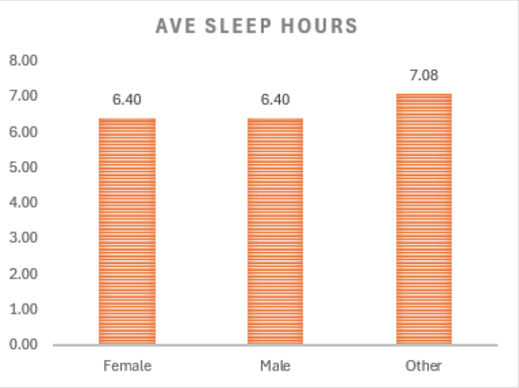
School Term

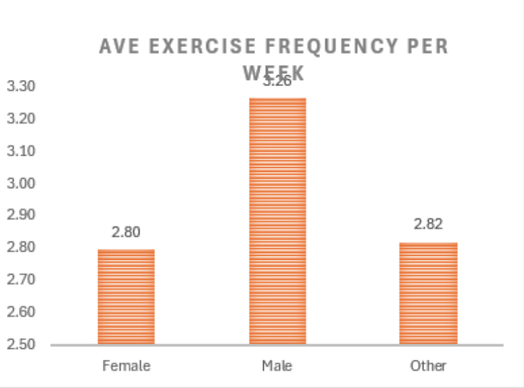
Semester 1 | Semester 2 | Summer Term

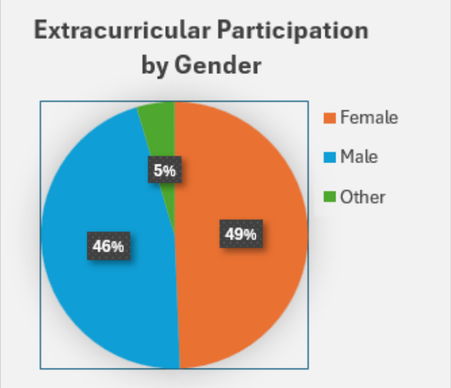
Gender

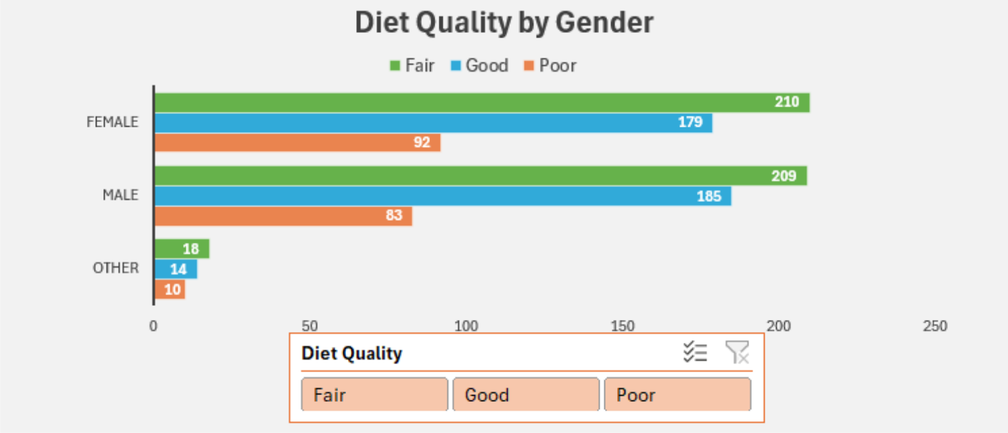
Female | Male | Other





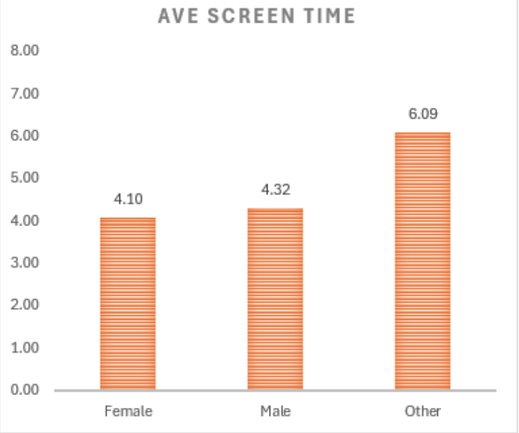


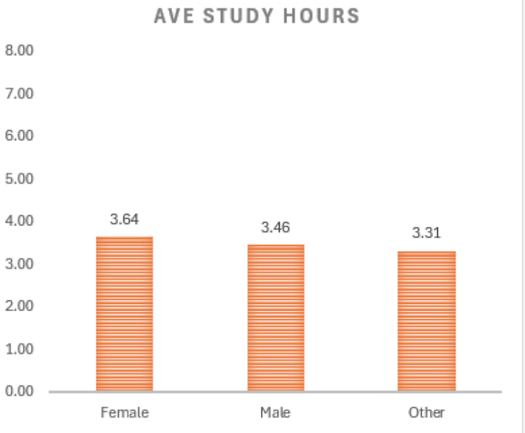


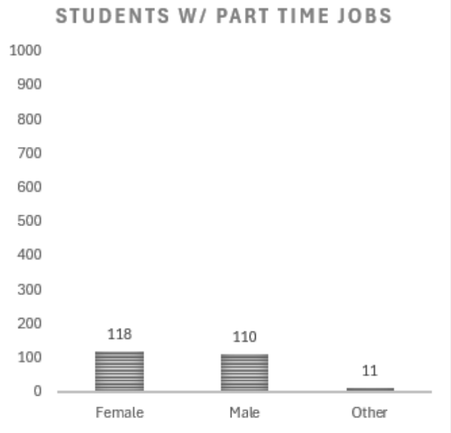


Diet Quality

Fair | Good | Poor







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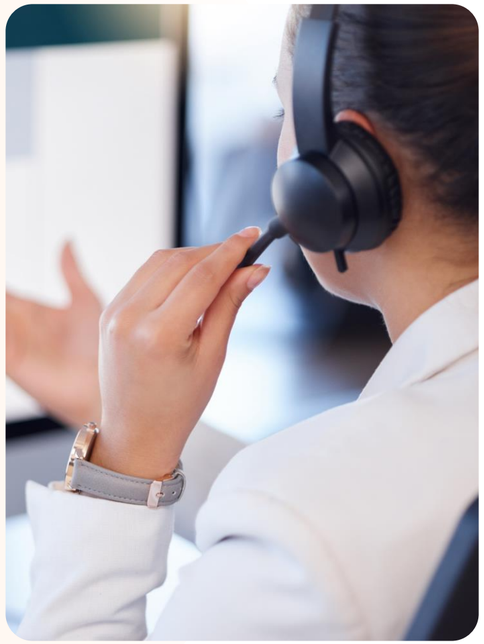
Thank You



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